

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

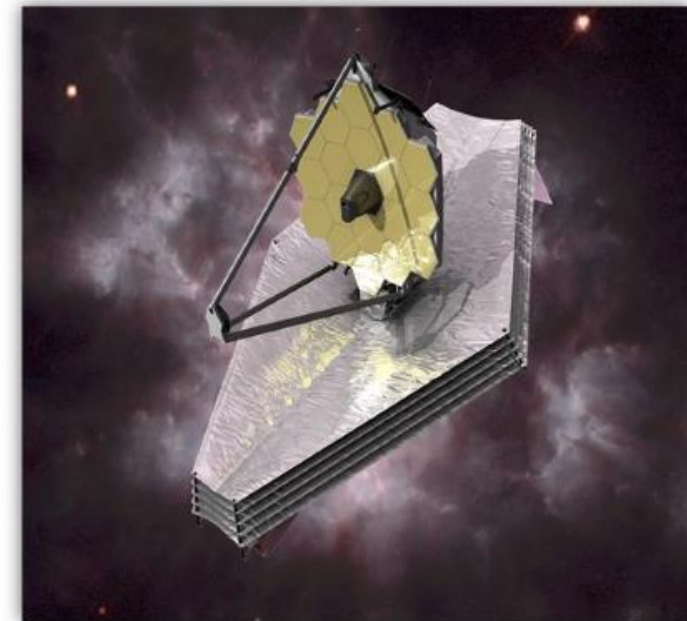
MODULE - 9

MAJOR MISSIONS OF OTHER COUNTRIES

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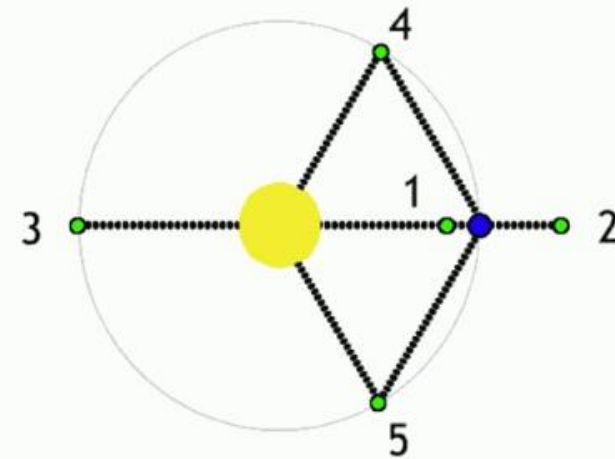
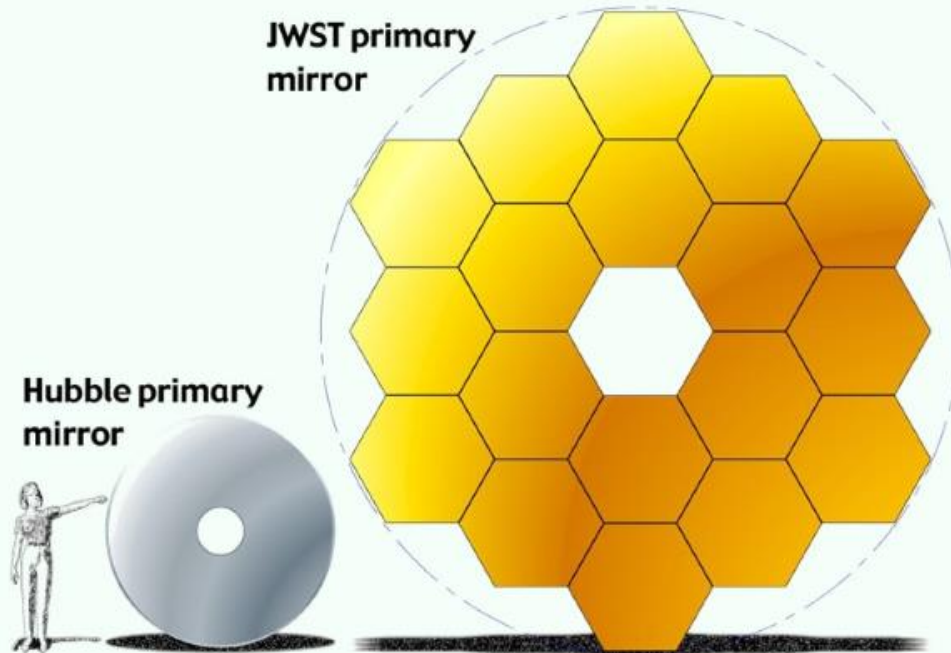
JAMES WEBB SPACE TELESCOPE

- JWST is a space telescope that will be the successor of Hubble Space Telescope.
- It will be launched in October 2021.
- The major goal is to observe some of the most distant objects in the universe, such as the first galaxies which are beyond the reach of the current instruments.
- JWST can detect both infrared and optical radiation.
- Its primary mirror is 6.5 meters, much larger than the primary mirror of HST which is 2.4 meters. So JWST provides much higher resolution than HST.
- JWST will be deployed at L2 Lagrange Point.



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JAMES WEBB SPACE TELESCOPE



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NASA-ISRO SYNTHETIC APERTURE RADAR

It is a joint project between NASA and ISRO to develop and launch a Synthetic Aperture RADAR (SAR).

WHAT IS A SAR?

- SAR is a type of RADAR that is used to create 2 or 3 dimensional images of objects such as landscapes.
- SAR has applications in:
 - ✓ Remote sensing
 - ✓ mapping of surfaces of earth and other planets
 - ✓ Oceanography
 - ✓ Glaciology
 - ✓ Forestry, etc.
- It is also used for military surveillance.



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- SAR has high resolution capacity independent of weather and day or night.
- NISAR is designed to observe and study processes like collapse of ice sheets in polar regions and natural hazards like earthquakes, tsunamis, volcanic eruptions etc.
- The satellite is planned to be launched in September 2022. It will be placed in a polar orbit.
- The satellite will have a mission life of 3 years.



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ARTEMIS PROGRAMME

- The Artemis program is NASA's human spaceflight program that has the goal of landing 'the first woman and the next man' on the Moon (lunar south pole region) by 2024.
- The program is carried out predominantly by NASA, U.S. commercial spaceflight companies contracted by NASA, and international partners including
 - the European Space Agency (ESA)
 - the Japan Aerospace Exploration Agency (JAXA)
 - Canadian Space Agency (CSA)
 - the Italian Space Agency (ASI)
 - the Australian Space Agency (ASA)
 - the UK Space Agency (UKSA) and
 - the United Arab Emirates Space Agency (UAESA).



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ARTEMIS PROGRAMME

- The program is divided into three parts:
 - ✓ Artemis I is most likely to be launched 2021 and involves an uncrewed flight to test the Space Launch System (SLS) and Orion spacecraft.
 - ✓ Artemis II will be the first crewed flight test and is targeted for 2023.
 - ✓ Artemis III will land astronauts on the Moon's South Pole in 2024.



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